Int J Fract (2026) 250:19
https://doi.org/10.1007/s10704-026-00913-2

RESEARCH

The Virtual Element Method for dynamic crack analysis

Philipp Wappler · **Kevin Schmitz** · **Andreas Ricoeur**

Received: 13 November 2025 / Revised: 2 February 2026 / Accepted: 23 February 2026
© The Author(s) 2026

Abstract The Virtual Element Method (VEM) constitutes a generalization of the standard Finite Element Method (FEM), allowing for substantially more flexibility in the mesh design. This feature makes the method particularly suitable for problems with evolving discontinuities and singularities, and thus for crack problems. Dynamic crack analyses oftentimes incorporate impact loads, requiring sophisticated procedures of time integration, which are known from the FEM. Furthermore, the crack tip loading quantities are highly dependent on structural damping, while algorithmic damping is inevitable for numerical stability. Established approaches to quantify crack tip loading are basically applicable within the VEM framework, however, have to be adapted to characteristics like the absence of shape functions within elements. The Displacement Interpretation Method (DIM), the Modified Crack Closure Integral and the J-integral are elaborated in this context, restricting analyses to linear elastic fracture mechanics and stationary cracks. The accuracy of the approaches is verified comparing results of analytical and other numerical calculations. Numerical studies reveal the impact of physical parameters associated with structural damping and load rate, as well as of numerical factors such as approximation order, contour of the J-integral, extrapolation domain of the DIM and algorithmic damping. After all, the VEM appears to be a highly suitable numerical approach for dynamic fracture mechanics with a view to future crack growth simulations.

Keywords Virtual Element Method · Elastodynamics · Dynamic fracture mechanics · Rayleigh damping

Institute of Mechanics, University of Kassel, Kassel 34125, Germany

1 Introduction

The accurate calculation of crack tip loading quantities in terms of stress intensity factors (SIFs), energy release rates or the J-integral is prerequisite for the simulation of crack propagation in natural or engineering structures. They essentially control rates of crack growth, crack deflection angles and crack path stability. While most work on fracture mechanics focusses on static problems, inertia must inevitably be incorporated in problems with mechanical impact, high-frequency loads or thermal shock, and may lead to dynamic overshoot or resonance-induced exaggeration of SIFs. Structural damping is inseparably linked to dynamic calculations, potentially reducing stress maxima and fluctuations due to excitation of higher order modes.

The Virtual Element Method (VEM) constitutes a generalization of the standard Finite Element Method (FEM), allowing for substantially more flexibility in the mesh design. Hanging nodes are allowed, just as non-convex elements which may even include crack tips. The discretization of crack problems is thus very efficient, inter alia, since the coarsening of the mesh

Published online: 29 March 2026

away from the crack tip or faces is hardly bound by any rules. In crack growth simulations, the crack may cut through elements, and remeshing is thus avoided without the need for enrichment of element shape functions. The calculation of crack tip loading quantities with the VEM, in particular in the context of dynamic fracture mechanics, however, is not state of the art.

A wide range of numerical methods has already been applied in the field of dynamic fracture mechanics. The Finite Difference Method (FDM) represents one, but rarely used, numerical approach that has been implemented for the analysis of dynamically loaded stationary and moving cracks, e.g., in Shmuely and Alterman (1973); Chen (1975) and Perl et al. (1982). The Boundary Element Method (BEM) has proven particularly suitable for various fracture applications, especially in linear elastic fracture mechanics (LEFM). A comprehensive overview is provided by Aliabadi (1997). In the context of dynamic fracture, e.g., the works of Chirino and Dominguez (1989) or Wen et al. (1996) are referred, which used Fourier and Laplace transformations to determine SIFs for crack loading analyses. Fedelinski et al. (1997) demonstrated the Dual Boundary Element Method (DBEM) in the time domain for rapid dynamic crack growth.

Probably the most prominent numerical method in engineering science is the FEM, which has become established, with various numerical fracture analysis techniques being developed. For instance, Nishioka and Atluri (1983a) and Nakamura et al. (1985) presented implementations of conservation integrals for dynamic processes. Jih and Sun (1990) applied the crack closure integral to determine time-dependent energy release rates. Enderlein et al. (2003) compared different FEM techniques in two and three dimensions. In addition to standard formulations, significant extensions to the FEM have been developed that effectively address the aforementioned numerical aspects, such as remeshing and hanging nodes. In this context, e.g., the works of Rashid (1998); Gürses and Miehe (2009); Kus et al. (2014) and Huo et al. (2020) can be referred to. The most important extension in the field of fracture mechanics is clearly the XFEM, whose foundations were laid by the works of Belytschko and Black (1999) and Moës et al. (1999). The approach addresses numerical limitations of FEM techniques for crack propagation simulations by enriching the classical FEM ansatz space, and has been applied in a wide range of works. A review paper of the XFEM is provided, e.g., by Sinha and Pamnani (2025).

Methods treating arbitrarily shaped elements are becoming increasingly common in numerical mechanics due to their high mesh flexibility. For example, the Polygonal Finite Element Method (PFEM), the Voronoi Cell Finite Element Method (VCFEM), or the Smoothed Finite Element Method (SFEM) can be categorized in this class of methods. The Scaled Boundary Finite Element Method (SBFEM) should also be mentioned here, which is based on the works of Song and Wolf (1996, 1997). For dynamic fracture problems, the semi-analytical scheme was applied, e.g., by Yang (2006) or Zhang et al. (2021). The VEM was developed by Beirão da Veiga et al. (2013a, 2014) and also belongs to the family of polytopal discretization schemes. The method evolved from the Mimetic Finite Difference Method (MFDM) and achieves the aforementioned mesh flexibility by defining the approximation functions only implicitly. Furthermore, projection operators on each element provide polynomial and consistent approximations of these functions, which allows partial differential equations to be solved with optimal numerical convergence rates.

The VEM for linear elasticity was presented, e.g., by Beirão da Veiga et al. (2013b) and Artioli et al. (2017). Applications of the method to nonlinear elasticity are found for instance in Wriggers and Hudobivnik (2017) and Liguori et al. (2024). Problems of elastodynamics were addressed by Park et al. (2019); Cihan et al. (2021) and Antonietti et al. (2021). General applications of the VEM in the field of mathematics and engineering are found in Antonietti (2022); Beirão da Veiga et al. (2023) and Wriggers et al. (2024).

The freedom in element design offered by the VEM was also exploited for problems in fracture mechanics. Nguyen-Thanh et al. (2018) and Hussein et al. (2019) were the first to use the VEM for crack tip loading analysis and quasi-static crack propagation in brittle materials. Marfia et al. (2022) and Choi et al. (2023) implemented cohesive zone models in a VEM framework. Chen et al. (2025) use a J-integral-based fracture criterion for quasi-static crack propagation. For dynamic fracture processes, the VEM has so far only been applied to phase field models, see, e.g., Aldakheel et al. (2019); Leng et al. (2025) and Xu et al. (2025). Crack analysis within an elastodynamic framework of LEFM has, to our best knowledge, not yet been realized with the VEM.

This paper is structured as follows: Sec. 2 introduces the strong and weak form of of the elastodynamic initial boundary value problem (IBVP), on which the subsequent loading analysis is based. Sec. 3 presents the mathematical framework and the spatial discretization with the VEM. Sec. 4 addresses the time discretization with the implicit Generalized Newmark Methods (GNMs), while in Sec. 5 the Displacement Interpretation Method (DIM), the Modified Crack Closure Integral (MCCI) and the J-integral for crack tip loading analyses of stationary cracks are outlined against the background of a VEM implementation. Finally, Sec. 6 contains three numerical examples in which time-dependent fracture mechanics quantities are calculated and compared to literature as well as to FE simulations.

2 Problem statement

In order to analyze dynamically loaded cracks based on the fracture mechanics quantities presented in Sec. 5, a requirement is to solve the IBVP of elastodynamics (Kuna 2013; Gross and Seelig 2018). To define the problem, a three-dimensional elastic body in the time interval $I_t = [t_0, T]$ with $T > t_0$ is considered. The body has a constant thickness B and is in a plane strain or plane stress state. For this work, the IBVP is formulated in two dimensions using the mid-surface representation of the body, as shown in Fig. 1.

For the mathematical description of the two-dimensional case, the domain of the mid-surface is denoted by $\Omega \subset \mathbb{R}^2$ with boundary $\partial\Omega = \Gamma$. The crack is indicated by $\Gamma^+ \subset \Gamma$ and $\Gamma^- \subset \Gamma$. Moreover, an arbitrary time-dependent surface load $\mathbf{t} = \bar{\mathbf{t}}$ is applied to the Neumann boundary $\Gamma_\sigma \subset \Gamma$, while inhomogeneous displacement values $\mathbf{u} = \bar{\mathbf{u}}$ are specified on the Dirichlet boundary $\Gamma_\mathbf{u} \subset \Gamma$. Under the assumption of small deformations, the strain–displacement relation is given by

$$\boldsymbol{\epsilon} = \frac{1}{2}\left(\nabla\mathbf{u} + (\nabla\mathbf{u})^\top\right) . \tag{1}$$

The stresses are related to the strains by the linear constitutive law

$$\boldsymbol{\sigma}(\mathbf{u}) = \mathbf{C} : \boldsymbol{\epsilon} = \mathbf{C} : \nabla\mathbf{u} . \tag{2}$$

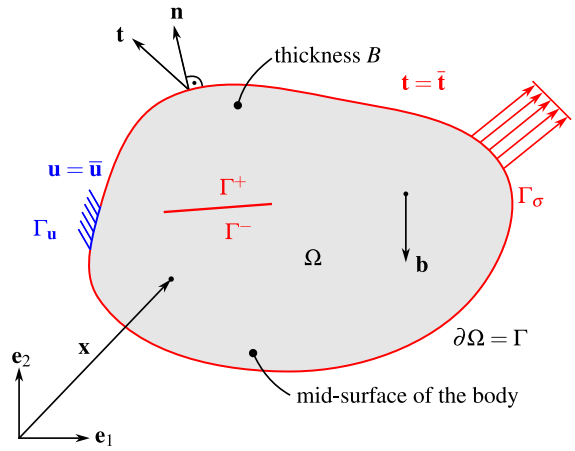

Fig. 1 Elastodynamic initial boundary value problem

Assuming an elastic isotropic material behavior, the elasticity tensor is defined by

$$\mathbf{C} = 2\mu\mathbb{I} + \tilde{\lambda}\mathbf{1} \otimes \mathbf{1} , \tag{3}$$

with the second-rank, respectively, fourth-rank identity tensors $\mathbf{1}$ and $\mathbb{I}$. For a plane strain or plane stress state the auxiliary constant (Hughes 2000; Slaughter 2002)

$$\tilde{\lambda} = \begin{cases} \lambda & \text{, plane strain} \\ \dfrac{2\mu\lambda}{\lambda + 2\mu} & \text{, plane stress} \end{cases} \tag{4}$$

is introduced, where λ and μ represent the classical Lamé constants. In addition to Eqs. (1) and (2), the balance law

$$\rho\ddot{\mathbf{u}} = \nabla \cdot \boldsymbol{\sigma}(\mathbf{u}) + \mathbf{b} \tag{5}$$

defines the equation of motion for the elastic body in terms of a second order hyperbolic partial differential equation, where ρ denotes the mass density and $\mathbf{b}$ the body forces. Introducing $\mathbf{x}$ as the position vector, the initial values

$$\mathbf{u}(\mathbf{x}, t_0) = \mathbf{u}_0 \tag{6}$$

$$\dot{\mathbf{u}}(\mathbf{x}, t_0) = \dot{\mathbf{u}}_0 \tag{7}$$

for displacement and velocity must be specified.

Analogous to the FEM, the VEM is a Bubnov-Galerkin method which requires a weak formulation of

the IBVP for a spatial discretization (Mengolini et al. 2019; Beirão da Veiga et al. 2013b). Let $\delta\mathbf{u} \in \mathbf{U}(\Omega) := \mathbf{H}^1(\Omega)$ be an arbitrary time-independent test function, which satisfies homogeneous Dirichlet conditions and $\mathbf{H}^1(\Omega) := \left[H^1(\Omega)\right]^2$ be the vector-valued first-order Sobolev space, the weak form can then be expressed as follows:

Find $(\mathbf{u}(t))(\mathbf{x}) = \mathbf{u}(\mathbf{x}, t) \in \mathbf{U}(\Omega)$ such that

$$\delta \mathrm{W}_{\mathrm{dyn}}(\delta\mathbf{u}, \mathbf{u}) + \delta \mathrm{W}_{\mathrm{int}}(\delta\mathbf{u}, \mathbf{u}) = \delta \mathrm{W}_{\mathrm{ext}}(\delta\mathbf{u}) \tag{8}$$

with

$$\delta \mathrm{W}_{\mathrm{dyn}}(\delta\mathbf{u}, \mathbf{u}) = B \int_{\Omega} \rho \delta\mathbf{u} \cdot \ddot{\mathbf{u}} \, \mathrm{d}A \tag{9}$$

$$\delta \mathrm{W}_{\mathrm{int}}(\delta\mathbf{u}, \mathbf{u}) = B \int_{\Omega} \nabla\delta\mathbf{u} : \mathbf{C} : \nabla\mathbf{u} \, \mathrm{d}A \tag{10}$$

$$\delta \mathrm{W}_{\mathrm{ext}}(\delta\mathbf{u}) = B \int_{\Gamma_\sigma} \delta\mathbf{u} \cdot \bar{\mathbf{t}} \, \mathrm{d}S + B \int_{\Omega} \delta\mathbf{u} \cdot \mathbf{b} \, \mathrm{d}A \tag{11}$$

and

$$\int_{\Omega} \delta\mathbf{u} \cdot (\mathbf{u}(\mathbf{x}, t_0) - \mathbf{u}_0) \, \mathrm{d}A = 0 \tag{12}$$

$$\int_{\Omega} \delta\mathbf{u} \cdot (\dot{\mathbf{u}}(\mathbf{x}, t_0) - \dot{\mathbf{u}}_0) \, \mathrm{d}A = 0 \tag{13}$$

is satisfied for all $\delta\mathbf{u} \in \mathbf{U}(\Omega)$. Eqs. (9) - (11) represent the virtual works for the elastic body. Finally, Eqs. (12) and (13) define the initial conditions in a weak form.

3 Spatial discretization with the Virtual Element Method

3.1 VEM discretization

This section introduces the VEM function spaces and the mathematical framework for approximating the displacement field of the presented IBVP. For the discretization, this work focuses on approximations of order $k \leq 2$, using continuous displacement interpolation, where $k \in \mathbb{N}$ denotes the approximation order. The outline of the method in this Section follows mainly the original works of Beirão da Veiga et al. (2013a, b), and Beirão da Veiga et al. (2014). In addition, the contributions of Mengolini et al. (2019); Artioli et al. (2017); Antonietti et al. (2021) and Wriggers et al. (2024) are addressed, who applied the method for elastic problems.

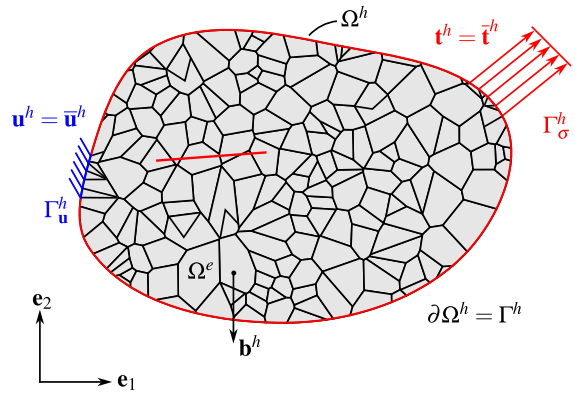

Fig. 2 Discretization of the domain

To achieve a displacement solution, the spatial domain is divided into a set of pairwise non-overlapping elements $\Omega^e \subset \Omega^h \approx \Omega$. The parameter $h \in \mathbb{R}^+$ represents the average element diameter for the partition and labels all numerical quantities that depend on the discretization. As illustrated in Fig. 2, any arbitrary convex and non-convex element shapes can be chosen for the mesh, while hanging nodes are also supported.
The solution space for the discretization is approximated by the global VEM space $\mathbf{U}^h_k(\Omega^h) \subset \mathbf{U}(\Omega)$. Similar to the FEM formulation, the global space is obtained through continuous assembly of the local element spaces $\mathbf{U}^h_k(\Omega^e) \subset \mathbf{U}^h_k(\Omega^h)$. The local definition will be specified more precisely in the following discussion. For this purpose the polynomial space on the two-dimensional virtual element, indicated by $\mathbb{P}_k(\Omega^e)$, is introduced. As basis functions, the scaled monomials

$$\xi^l = \left(\frac{x - \bar{x}^e}{h^e}\right)^l, \qquad \eta^l = \left(\frac{y - \bar{y}^e}{h^e}\right)^l, \qquad l = 1, \ldots, k$$

are utilized. Here, $\mathbf{x}^e_C = \left[\bar{x}^e, \bar{y}^e\right]^\top \in \mathbb{R}^2$ represents the centroid coordinates and $h^e \in \mathbb{R}^+$ denotes the element diameter. For two-dimensional applications, the polynomial space is extended by $\mathbf{P}_k(\Omega^e) := \left[\mathbb{P}_k(\Omega^e)\right]^2$. This vector-valued space has the dimension $\dim \mathbf{P}_k(\Omega^e) = (k+1)(k+2) := N_k$, and the following basis, depending on the approximation order, is chosen:

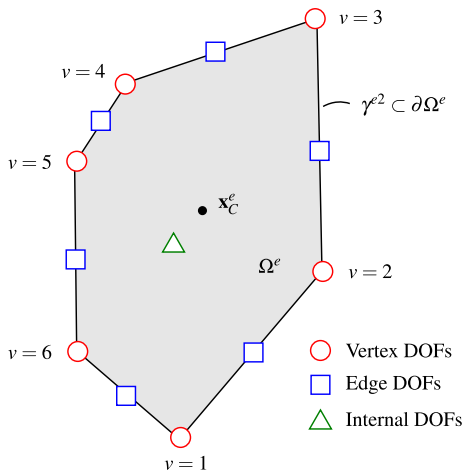

Fig. 3 Virtual element for approximation order of $k = 2$

$\underline{k = 1}$:

$$\chi_1 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -\eta \\ \xi \end{bmatrix}, \begin{bmatrix} \eta \\ \xi \end{bmatrix}, \begin{bmatrix} \xi \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ \eta \end{bmatrix} \right\} ,$$

$\underline{k = 2}$:

$$\chi_2 = \chi_1 \cup \left\{ \begin{bmatrix} \xi^2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ \xi^2 \end{bmatrix}, \begin{bmatrix} \xi\eta \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ \xi\eta \end{bmatrix}, \begin{bmatrix} \eta^2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ \eta^2 \end{bmatrix} \right\} .$$

Following counter-clockwise vertex numbering by the index $v \in \{1, \ldots, N_V^e\}$ for $N_V^e \in \mathbb{N}^{\geq 3}$ vertices per element, the connecting edges are denoted by $\gamma^{ev} \subset \partial\Omega^e$, as illustrated in Fig. 3. The corresponding local VEM space then comprises functions $\mathbf{w}^h(\mathbf{x}, \tau) \in \mathbf{U}_k^h(\Omega^e)$ whose components, for a fixed point in time $\tau \in I_t$, satisfy the following properties (Beirão da Veiga et al. 2013a):

Property 1: $w_i^h(\mathbf{x}, \tau)$ is continuous on the element boundary $\partial\Omega^e$,
Property 2: $w_i^h(\mathbf{x}, \tau)$ is a polynomial with degree $k \in \mathbb{N}$ on every edge $\gamma^{ev} \subset \partial\Omega^e$,
Property 3: $\Delta w_i^h(\mathbf{x}, \tau)$ is a polynomial of degree $k-2$ in the interior of the virtual element.

A key difference in the definition of the virtual element functions, as opposed to finite element functions, is that they are only implicitly defined on the element. The advantage of this approach is a greater flexibility in the choice of element shapes, which enables the versatility of the VEM in respect of discretization. The presented functional properties lead to the mathematical definition

$$\begin{aligned} \mathbf{U}_k^h(\Omega^e) = \Big\{ & \mathbf{w}^h \in \left[H^1(\Omega^e) \cap C^0(\Omega^e) \right]^2 : \\ & w_i^h|_{\partial\Omega^e} \in C^0(\partial\Omega^e),\ w_i^h|_{\gamma^{ev} \subset \partial\Omega^e} \in \mathbb{P}_k(\gamma^{ev}), \\ & \Delta w_i^h|_{\Omega^e} \in \mathbb{P}_{k-2}(\Omega^e),\ \forall\, v = 1, \ldots, N_V^e,\ \forall\, i = 1, 2 \Big\} . \end{aligned}$$

The dimension for this space is $\dim \mathbf{U}_k^h(\Omega^e) = 2N_V^e k + (k-1)k := N_D^e$, representing the number of degrees of freedom that characterize a virtual element function. For the local VEM space, it should be noted that the relation $\mathbf{P}_k(\Omega^e) \subset \mathbf{U}_k^h(\Omega^e)$ holds. Let $A^e \in \mathbb{R}^+$ denote the element area. The following degrees of freedom are chosen for the VEM formulation, as shown in Fig. 3 (Mengolini et al. 2019):

Vertex-DOFs: $(i = 1, \ldots, 2N_V^e)$ For $k \geq 1$ the values at the vertices.
Edge-DOFs: $(i = 2N_V^e + 1, \ldots, 4N_V^e)$ For $k \geq 2$ the values at the $k-1$ points along each edge.
Internal-DOFs: $(i = 4N_V^e + 1, \ldots, N_D^e)$ For $k \geq 2$ the moments with

$$\frac{1}{A^e} \int_{\Omega^e} \mathbf{w}^h \cdot \mathbf{p} \, \mathrm{d}A \qquad \forall\, \mathbf{p} \in \mathbf{P}_{k-2}(\Omega^e) .$$

In contrast to the FEM, a second or higher order approximation requires additional internal degrees of freedom, called moments. In this work, the moment is equivalent to the integral mean of the displacement on an element.

With the definitions established so far, a semi-discrete displacement ansatz can finally be formulated. Denoting

$$\Lambda_k = \left\{ \mathbf{N}_1, \ldots, \mathbf{N}_{N_D^e} \right\}$$

as basis of the k-th order local VEM space, the semi-discrete ansatz is

$$\mathbf{u}^h(\mathbf{x}, t) = \sum_{i=1}^{N_D^e} u^{ei}(t) \mathbf{N}_i(\mathbf{x}) , \tag{14}$$

and can be used to interpolate the unknown displacement field. For inhomogeneous Dirichlet boundary conditions, the discrete values of constrained degrees of freedom must be incorporated into the formulation.

3.2 Implementation of the projection operator

A key concept for the spatial discretization with the VEM is the use of projection operators. The operators enable the mapping between the VEM function space $\mathbf{U}_k^h(\Omega^e) \subset \mathbf{U}(\Omega^e)$ and the vector valued polynomial space $\mathbf{P}_k(\Omega^e) \subset \mathbf{U}_k^h(\Omega^e)$, while preserving the approximation accuracy and the convergence rate of the method. The reason why operators are introduced lies in the fact that VEM functions are implicitly defined by the properties outlined in Sec. 3.1 and can't be used directly. The projection provides an optimal polynomial approximation for the displacement field to calculate element quantities. In the literature, different operators are defined for various types of problems in the field of mathematics and engineering science (Wriggers et al. 2024; Antonietti 2022; Dassi 2025). For applications of the VEM in case of linear elastodynamics, the operator

$$\begin{aligned} \Pi &: \mathbf{U}_k^h(\Omega^e) \to \mathbf{P}_k(\Omega^e) \\ \mathbf{u}^h &\mapsto \Pi(\mathbf{u}^h) := \mathbf{u}_\pi^h \end{aligned}$$

is used. To construct this projection operator, the discrete bilinear form

$$Z_{\text{int}}^e(\mathbf{v}^h, \mathbf{u}^h) = B \int_{\Omega^e} \nabla \mathbf{v}^h : \mathbf{C} : \nabla \mathbf{u}^h \, \mathrm{d}A \tag{15}$$

is introduced as a generalization of Eq. (10), where $\mathbf{v} \in \mathbf{U}_k^h(\Omega^e)$ denotes a test function. Applied to a single element, the operator satisfies the Galerkin orthogonality condition (Mengolini et al. 2019; Sukumar and Tupek 2022)

$$Z_{\text{int}}^e(\mathbf{p}, \mathbf{u}^h - \mathbf{u}_\pi^h) = 0 \quad \forall\, \mathbf{u}^h \in \mathbf{U}_k^h(\Omega^e), \;\; \mathbf{p} \in \mathbf{P}_k(\Omega^e) \,. \tag{16}$$

This projection essentially approximates the gradient of a VEM function, and thereby enables the calculation of an element stiffness matrix. Furthermore, Ahmad et al. (2013) demonstrate that the operator provides an L^2-projection in the case of $k \leq 2$. If the discrete bilinear form

$$Z_{\text{dyn}}^e(\mathbf{v}^h, \mathbf{u}^h) = B \int_{\Omega^e} \rho \mathbf{v}^h \cdot \ddot{\mathbf{u}}^h \, \mathrm{d}A \,, \tag{17}$$

associated with Eq. (9) of the weak formulation, is defined for the approximation of the dynamic element work, the operator consequently fulfills the additional condition

$$Z_{\text{dyn}}^e(\mathbf{p}, \mathbf{u}^h - \mathbf{u}_\pi^h) = 0 \quad \forall\, \mathbf{u}^h \in \mathbf{U}_k^h(\Omega^e), \;\; \mathbf{p} \in \mathbf{P}_k(\Omega^e) \tag{18}$$

and can be used to determine element mass matrices and suitable load vectors. By using the defined operator, the projection of the semi-discrete ansatz in Eq. (14) is given by

$$\mathbf{u}_\pi^h = \Pi(\mathbf{u}^h) = \sum_{i=1}^{N_D^e} u^{ei} \Pi(\mathbf{N}_i) = \sum_{i=1}^{N_D^e} \sum_{\alpha=1}^{N_k} u^{ei} s_i^\alpha \mathbf{p}_\alpha \,, \tag{19}$$

where the coefficients $s_i^\alpha = (\mathbf{\Pi})_{\alpha i}$ are unknown. Here, the matrix $\mathbf{\Pi} \in \mathbb{R}^{N_k \times N_D^e}$ denotes the representation of the projection operator with respect to the polynomial basis χ_k.

The VEM procedure requires the computation of the operator in matrix representation for each element. Therefore, the condition in Eq. (16) is applied, which leads to the algebraic equation

$$\mathbf{\Pi} = \mathbf{G}^{-1}\mathbf{B} \,. \tag{20}$$

For the matrices $\mathbf{G} \in \mathbb{R}^{N_k \times N_k}$ and $\mathbf{B} \in \mathbb{R}^{N_k \times N_D^e}$ the definitions

$$(\mathbf{G})_{\alpha\beta} = \begin{cases} \int_{\partial\Omega^e} \mathbf{p}_\alpha \cdot \mathbf{p}_\beta \, \mathrm{d}S & \text{for } \alpha = 1, 2, 3 \\ Z_{\text{int}}^e(\mathbf{p}_\alpha, \mathbf{p}_\beta) & \text{for } \alpha = 4, \ldots, N_k \end{cases} \tag{21}$$

and

$$(\mathbf{B})_{\alpha i} = \begin{cases} \int_{\partial\Omega^e} \mathbf{p}_\alpha \cdot \mathbf{N}_i \, \mathrm{d}S & \text{for } \alpha = 1, 2, 3 \\ Z_{\text{int}}^e(\mathbf{p}_\alpha, \mathbf{N}_i) & \text{for } \alpha = 4, \ldots, N_k \,. \end{cases} \tag{22}$$

derive from Eq. (16). Due to the fact that the projection condition defines the presented matrices only up to the first three rows, additional linear independent equations have to be added. For this purpose, the condition

$$\int_{\partial\Omega^e} \mathbf{p}_\alpha \cdot \left(\mathbf{u}^h - \mathbf{u}_\pi^h\right) \mathrm{d}S = 0 \tag{23}$$

for $\alpha = 1, 2, 3$ is adopted (Ahmad et al. 2013). The matrix $\mathbf{G}$ can then be computed exactly by integration. For this work, the standard Gauss-Lobatto rule is used to evaluate the boundary integrals. To integrate polynomials over polygonal elements, the scaled boundary cubature scheme developed by Chin and Sukumar (2021) is employed. Similarly, the matrix $\mathbf{B}$ is computed exactly with a boundary integration for $\alpha = 1, 2, 3$. For $\alpha > 3$ its definition requires a manipulation using the product rule and Gauss's theorem. This leads to the expression

$$(\mathbf{B})_{\alpha i} = Z^e_{\text{int}}(\mathbf{p}_\alpha, \mathbf{N}_i) = B \int_{\partial\Omega^e} \mathbf{n} \cdot (\mathbf{N}_i \cdot \mathbf{C} : \nabla \mathbf{p}_\alpha) \, \mathrm{d}S - B \int_{\Omega^e} \mathbf{N}_i \cdot \nabla \cdot (\mathbf{C} : \nabla \mathbf{p}_\alpha) \, \mathrm{d}A \, . \tag{24}$$

The first term is evaluated by using again the Gauss-Lobatto rule. The second term is computed based on the definition for the internal degrees of freedom. This is possible because the relation $\nabla \cdot (\mathbf{C} : \nabla \mathbf{p}_\alpha) \in \mathbf{P}_{k-2}(\Omega^e)$ holds.

For the application of VEM stabilization techniques, see Secs. 3.3 and 3.4, it is necessary to express the projection operator with respect to the basis Λ_k, which is denoted by $\tilde{\mathbf{\Pi}} \in \mathbb{R}^{N^e_D \times N^e_D}$. Introducing the DOFI-operator, which evaluates a function at the i-th degree of freedom of a virtual element, the matrix

$$(\mathbf{L})_{i\alpha} = \mathrm{dof}_i(\mathbf{p}_\alpha) \tag{25}$$

enables the computation. It can be shown that

$$\tilde{\mathbf{\Pi}} = \mathbf{L}\mathbf{\Pi} \tag{26}$$

holds. Finally, the relation

$$\mathbf{G} = \mathbf{B}\mathbf{L} \tag{27}$$

is introduced. This equation, e.g., proven by Beirão da Veiga et al. (2014), is used to save numerical effort, since computing the matrix $\mathbf{G}$ is not explicitly required to obtain the operator representations. However, as e.g. Mengolini et al. (2019) point out, the matrix $\mathbf{G}$ can be used to verify implementations.

3.3 Implementation of the element stiffness matrix

Computing the element stiffness matrix is straightforward, if the matrix representation of the projection operator is available. For this purpose the VEM basis function is rewritten in term of a projection and a remainder part as (Beirão da Veiga et al. 2014)

$$\mathbf{N}_i = \underbrace{\Pi(\mathbf{N}_i)}_{\text{projection}} + \underbrace{(\mathbf{N}_i - \Pi(\mathbf{N}_i))}_{\text{remainder}} \, . \tag{28}$$

Substituting this extension into Eq. (15) the mixed terms vanish due to Eq. (16). The stiffness matrix is then composed of two terms with

$$(\mathbf{k}^e)_{ij} = Z^e_{\text{int}}(\mathbf{N}_i, \mathbf{N}_j) = (\mathbf{k}^e_c)_{ij} + (\mathbf{k}^e_s)_{ij} \, . \tag{29}$$

The first term, which corresponds to the projection of VEM basis functions, is called consistency part and is computed by

$$\mathbf{k}^e_c = \mathbf{\Pi}^\top \tilde{\mathbf{G}} \mathbf{\Pi} \, . \tag{30}$$

Here, the matrix $\tilde{\mathbf{G}} \in \mathbb{R}^{N_k \times N_k}$ is constructed by setting the first three rows of the matrix given in Eq. (21) to zero. The second term is the so-called stabilization part. This term corresponds to the remainder and is therefore not computable. At this point, the VEM procedure involves stabilization techniques, where the term is approximated according to the underlying problem. For problems in elasticity, the DOFI-stabilization

$$\mathbf{k}^e_s = \gamma \, \mathsf{tr}(\mathbf{k}^e_c) \left(\mathbf{I} - \tilde{\mathbf{\Pi}}\right)^\top \left(\mathbf{I} - \tilde{\mathbf{\Pi}}\right) \tag{31}$$

is frequently employed, which is also adopted for this work. Other stabilization options can be found, e.g., in Wriggers et al. (2024). In the formulation $\mathbf{I} \in \mathbb{R}^{N^e_D \times N^e_D}$ represents the identity matrix, while $\mathsf{tr}(\cdot)$ is the trace operator. Referring to Mengolini et al. (2019) the parameter $\gamma > 0$ is customizable and without loss of generality set to $\gamma = 0.5$ in the case of linear elasticity.

3.4 Implementation of the element mass matrix

The element mass matrix is calculated in analogous manner to the stiffness matrix. The VEM basis function

extension in Eq. (28) is substituted into Eq. (17) for each element. Similarly, the mixed terms disappear due to the additional condition in Eq. (18), resulting in

$$(\mathbf{m}^e)_{ij} = Z^e_{\text{dyn}}(\mathbf{N}_i, \mathbf{N}_j) = (\mathbf{m}^e_c)_{ij} + (\mathbf{m}^e_s)_{ij} \,. \tag{32}$$

The consistency term is obtained as

$$\mathbf{m}^e_c = \mathbf{\Pi}^\top \mathbf{A} \mathbf{\Pi} \,, \tag{33}$$

where the entries of the matrix $\mathbf{A} \in \mathbb{R}^{N_k \times N_k}$ with

$$(\mathbf{A})_{\alpha\beta} = Z^e_{\text{dyn}}(\mathbf{p}_\alpha, \mathbf{p}_\beta) \tag{34}$$

require an integration over the element. For the stabilization term, the approach presented by Park et al. (2019) is followed according to

$$\mathbf{m}^e_s = \rho A^e B \left(\mathbf{I} - \tilde{\mathbf{\Pi}}\right)^\top \left(\mathbf{I} - \tilde{\mathbf{\Pi}}\right) \,. \tag{35}$$

In contrast to the stiffness matrix, the mass density and the geometric element quantities scale the term.

3.5 Implementation of the load vector

Based on the discrete linear form

$$Z_{\text{ext}}(\mathbf{v}^h) = B \int\limits_{\Gamma^h_\sigma \cap \partial\Omega^e} \mathbf{v} \cdot \bar{\mathbf{t}}^h \, \mathrm{d}S + B \int\limits_{\Omega^e} \mathbf{v} \cdot \mathbf{b}^h \, \mathrm{d}A \,, \tag{36}$$

as a generalization of Eq. (11), and applied to a single element, the VEM load vector

$$(\mathbf{r}^e)_i = Z^e_{\text{ext}}(\mathbf{N}_i) = (\mathbf{r}^e_\sigma)_i + (\mathbf{r}^e_\mathbf{b})_i \tag{37}$$

is composed of two parts. The first part

$$(\mathbf{r}^\mathbf{e}_\sigma)_i = B \int\limits_{\Gamma^h_\sigma \cap \partial\Omega^e} \mathbf{N}_i \cdot \bar{\mathbf{t}}^h \, \mathrm{d}S \,. \tag{38}$$

involves the Neumann boundary condition and can be treated as in the FEM by performing a boundary integration with discrete values of the applied load function. For the body load part

$$(\mathbf{r}^e_\mathbf{b})_i \approx B \int\limits_{\Omega^e} \Pi(\mathbf{N}_i) \cdot \mathbf{b}^h \, \mathrm{d}A \tag{39}$$

only the projection of the VEM basis function inside the element can be used. However, Ahmad et al. (2013) showed that this approach provides an optimal approximation.

3.6 Assembly of matrices, computation of stresses and structural damping

Based on the element matrices, a global system is assembled in terms of second order, linear and inhomogeneous ordinary differential equations according to

$$\mathbf{M}\ddot{\mathbf{u}}(t) + \mathbf{K}\mathbf{u}(t) = \mathbf{f}(t) \,, \tag{40}$$

involving those degrees of freedom (DOFs) that are unaffected by Dirichlet boundary conditions. If N_{Dof} defines their numbers, the matrices $\mathbf{M}, \mathbf{K} \in \mathbb{R}^{N_{\text{Dof}} \times N_{\text{Dof}}}$ represent the associated global mass and stiffness matrices. The time dependent function $\mathbf{f}(t) : I_t \longrightarrow \mathbb{R}^{N_{\text{Dof}}}$ corresponds to the nodal forces, while $\mathbf{u}(t) : I_t \longrightarrow \mathbb{R}^{N_{\text{Dof}}}$ governs the displacement DOFs. For both functions, it is assumed that they are continuous with respect to time. For the subsequent time discretization, it is additionally assumed that the displacement vector is sufficiently often continuously differentiable and integrable. Together with prescribed discrete initial values, following Eqs. (6), (7) and, respectively, Eqs. (12), (13), the differential equations constitute a semidiscrete initial value problem (IVP), whose solution describes the time evolution of the displacements and is obtained by applying time discretization methods, see Sec. 4.

The stresses are computed from the displacements for each element in a postprocessing step and the gradient of the VEM basis function is approximated by its projection. Using the constitutive law in Eq. (2) the stresses are calculated by

$$\boldsymbol{\sigma} \approx \boldsymbol{\sigma}_\pi = \mathbf{C} : \nabla \mathbf{u}_\pi \,. \tag{41}$$

To take damping into account, a simple but often sufficient approach in structural mechanics is to replace the

constitutive relation in Eq. (2) by

$$\boldsymbol{\sigma} = \mathbf{C} : (\boldsymbol{\epsilon} + \alpha\dot{\boldsymbol{\epsilon}}) = \mathbf{C} : \nabla\,(\mathbf{u} + \alpha\dot{\mathbf{u}}) \ . \tag{42}$$

This modification leads to an additional term

$$\delta W_{\text{diss}}(\delta\mathbf{u}, \dot{\mathbf{u}}) = \alpha B \int_{\Omega} \nabla\delta\mathbf{u} : \mathbf{C} : \nabla\dot{\mathbf{u}} \, \mathrm{d}A \tag{43}$$

in Eq. (8), and to a velocity proportional Rayleigh damping force with damping matrix (Hughes 2000; Bathe 1996; Zienkiewicz and Taylor 2000)

$$\mathbf{D} = \alpha\mathbf{K} \tag{44}$$

introduced in Eq. (40). The coefficient $\alpha \in \mathbb{R}$ controls the global influence of the stiffness-proportional damping components.

4 Time discretization with the Generalized Newmark Methods (GNMs)

To solve the discrete IVP according to Sec. 3.6, the GNMs are employed. These methods form a family of implicit, single-step time integration schemes, based on the theoretical foundation introduced by Newmark (1959). Over the years the Newmark-based approaches have been further developed, where significant algorithmic extensions were made, e.g., by Chung and Hulbert (1993); Wood et al. (1980) and Hilber et al. (1977). The methods are accurate up to the second order and unconditionally stable for the considered system of linear differential equations. Furthermore, these methods have intrinsic numerical dissipation, which damps higher modes, depending on the dissipation parameter denoted by $\rho_\infty \in \mathbb{R}^+$.

The time domain is discretized into a set of non-overlapping discrete time intervals $\Delta t_n = t_{n+1} - t_n$. Referring to Newmark (1959), two linear acceleration approaches on a time interval are made, parameterized by numerical constants $\beta, \gamma \in \mathbb{R}$. Integrating these approaches, leads to the relations

$$\dot{\mathbf{u}}_{n+1} = \frac{\gamma}{\beta\Delta t_n}\left(\mathbf{u}_{n+1} - \mathbf{u}_n\right) - \frac{\gamma-\beta}{\beta}\dot{\mathbf{u}}_n - \frac{\gamma-2\beta}{2\beta}\Delta t_n\ddot{\mathbf{u}}_n \ , \tag{45}$$

$$\ddot{\mathbf{u}}_{n+1} = \frac{1}{\beta\Delta t_n^2}\left(\mathbf{u}_{n+1} - \mathbf{u}_n\right) - \frac{1}{\beta\Delta t_n}\dot{\mathbf{u}}_n - \frac{1-2\beta}{2\beta}\ddot{\mathbf{u}}_n \ . \tag{46}$$

The velocity and acceleration are computed based on the displacement solution $\mathbf{u}(t_{n+1}) = \mathbf{u}_{n+1}$ of the next time step. Following the approaches by Chung and Hulbert (1993) this displacement solution is determined by applying the equation of motion

$$\mathbf{M}\ddot{\mathbf{u}}_{n+1-\alpha_m} + \mathbf{D}\dot{\mathbf{u}}_{n+1-\alpha_f} + \mathbf{K}\mathbf{u}_{n+1-\alpha_f} = \mathbf{f}_{n+1-\alpha_f} \tag{47}$$

at the weighted midpoints with

$$\mathbf{u}_{n+1-\alpha_f} = (1-\alpha_f)\mathbf{u}_{n+1} + \alpha_f\mathbf{u}_n \ , \tag{48}$$

$$\dot{\mathbf{u}}_{n+1-\alpha_f} = (1-\alpha_f)\dot{\mathbf{u}}_{n+1} + \alpha_f\dot{\mathbf{u}}_n \ , \tag{49}$$

$$\ddot{\mathbf{u}}_{n+1-\alpha_m} = (1-\alpha_m)\ddot{\mathbf{u}}_{n+1} + \alpha_m\ddot{\mathbf{u}}_n \ , \tag{50}$$

$$\mathbf{f}_{n+1-\alpha_f} = (1-\alpha_f)\mathbf{f}_{n+1} + \alpha_f\mathbf{f}_n \ , \tag{51}$$

whereupon the constants $\alpha_f, \alpha_m \in \mathbb{R}$ control the evaluation of the quantities in the time interval. Mathematical manipulation of Eq. (47) leads to the linear system of equations

$$\mathbf{K}_n^{\text{eff}}\,\mathbf{u}_{n+1} = \mathbf{f}_n^{\text{eff}} \ , \tag{52}$$

which is characteristic for implicit time discretization methods. Solving Eq. (52) for every time step numerically, generates a discrete displacement solution. The velocity and acceleration are updated consistently by Eqs. (45) and (46). For an appropriate choice of the numerical parameters $\beta, \gamma, \alpha_f, \alpha_m$, depending on the dissipation parameter, the configurations proposed by Hilber et al. (1977) and Wood et al. (1980) are employed for this work:

<u>Hilber − α − Method</u> (Hilber et al. 1977) :
$\rho_\infty \in \left[\frac{1}{3}, 1\right]$

$$\beta = \frac{1}{4}\left(1+\alpha_f\right)^2 \ , \qquad \alpha_f = \frac{1-\rho_\infty}{\rho_\infty+1} \ ,$$
$$\gamma = \frac{1}{2} + \alpha_f \ , \qquad \alpha_m = 0 \ .$$

<u>Bossak − α − Method</u> (Wood et al. 1980):
$\rho_\infty \in [0, 1]$

$$\beta = \frac{1}{4}\left(1-\alpha_m\right)^2 \ , \qquad \alpha_f = 0 \ ,$$

$$\gamma = \frac{1}{2} - \alpha_m \,, \qquad \alpha_m = \frac{\rho_\infty - 1}{\rho_\infty + 1} \,.$$

5 Fracture mechanics quantities and numerical implementation

5.1 Displacement interpretation method (DIM)

For analyzing cracks under transient loads in a LEFM context, the nearfield solution for the displacement field is given, e.g., by Kuna (2013) or Gross and Seelig (2018) according to

$$\begin{bmatrix} u_1(r,\varphi,t) \\ u_2(r,\varphi,t) \end{bmatrix} = \frac{1}{2\mu}\sqrt{\frac{r}{2\pi}} \begin{bmatrix} g_1^I(\varphi) & g_1^{II}(\varphi) \\ g_2^I(\varphi) & g_2^{II}(\varphi) \end{bmatrix} \begin{bmatrix} K_I(t) \\ K_{II}(t) \end{bmatrix} \,, \tag{53}$$

whereupon static and dynamic solutions are basically identical for stationary cracks. The SIFs are explicitly time-dependent, while the angular functions are defined as in the static case by

$$g_1^I(\varphi) = \cos\left(\frac{\varphi}{2}\right)(\kappa - \cos(\varphi)) \,, \tag{54}$$

$$g_2^I(\varphi) = \sin\left(\frac{\varphi}{2}\right)(\kappa - \cos(\varphi)) \,, \tag{55}$$

$$g_1^{II}(\varphi) = \sin\left(\frac{\varphi}{2}\right)(\kappa + 2 + \cos(\varphi)) \,, \tag{56}$$

$$g_2^{II}(\varphi) = -\cos\left(\frac{\varphi}{2}\right)(\kappa - 2 + \cos(\varphi)) \,, \tag{57}$$

and the Kosolov constant by

$$\kappa = \begin{cases} 3 - 4\nu & \text{, plane strain} \\ \dfrac{3-\nu}{1+\nu} & \text{, plane stress} \end{cases} \,. \tag{58}$$

For the DIM the nearfield solution is evaluated in every time step by monitoring the numerical crack opening displacement (COD) at DOF pairs along the crack surfaces. The number of these pairs is denoted by N_{DIM}. The SIFs are determined by extrapolation, using a least squares method and linear extrapolation functions, including DOFs that provide a sufficiently accurate displacement solution on the one hand, and for which the $\sqrt{r}$-characteristic of the near field according to Eq. (53) constitutes a reasonable approximation on the other. To identify a suitable extrapolation domain valid for every time step, the COD was initially averaged over time, which saves numerical effort. In this context a time step weighted approach with

$$\langle \Delta u_j \rangle_t (r_i, \pm\pi) = \frac{1}{T} \sum_{n=0}^{N_T - 1} \Delta u_j (r_i, \pm\pi, t_{n+1}) \Delta t_n \tag{59}$$

for all $i = 1, \ldots, N_{\mathrm{DIM}}$ and displacement components $j = 1, 2$, is employed.

5.2 Modified crack closure integral (MCCI)

The dynamic energy release rate is derived by incorporating the kinetic energy K of the body, resulting in the definition (Kuna 2013; Gross and Seelig 2018)

$$G(t) = G_I(t) + G_{II}(t) = -\frac{1}{B}\frac{\mathrm{d}(U_{\mathrm{tot}} + K)}{\mathrm{d}a} \,, \tag{60}$$

where a denotes the crack length and U_{tot} represents the total potential energy. The energy release rate is formally split into contributions of Mode I and Mode II crack opening and can be calculated by the crack closure integral

$$G(t) = \lim_{\Delta a \to 0} \frac{1}{2B\Delta a} \int_0^{\Delta a} \sum_{i=1}^{2} \sigma_{2i}(s, 0, t) \Delta u_i (\Delta a - s, \pm\pi, t) \, \mathrm{d}s \,, \tag{61}$$

which includes the tractions σ_{2i} $(i = 1, 2)$ on the ligament of the initial crack and the COD Δu_i on the faces of the grown crack. Eq. (61) is basically established for static problems, however, it also applies to dynamic problems, since the specific crack closure work $\sigma_{2i}\Delta u_i$ implicitly accounts for work done against the inertia (Enderlein et al. 2003). To avoid calculations of two separate crack states, the modification of the crack closure integral proposed by Buchholz (1984) is adopted, whereupon just the initial crack is considered for integration. If the edge lengths L of the crack tip elements shown in Fig. 4 are kept constant, and are chosen sufficiently small in light of the required limit $\Delta a \to 0$, the energy release rate can be approximated using the near tip reaction forces at the VEM specific vertex and edge DOFs, resulting in Kuna (2013)

$$G_I(t) \approx \frac{1}{2BL} F_2^{\alpha}(t) \Delta u_2^{\alpha - 1}(t) \tag{62}$$

$$G_{II}(t) \approx \frac{1}{2BL} F_1^{\alpha}(t) \Delta u_1^{\alpha - 1}(t) \tag{63}$$

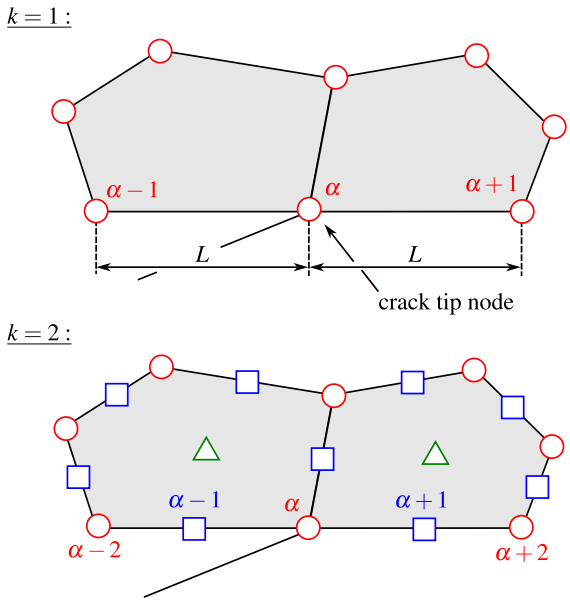

Fig. 4 Crack tip elements for the MCCI depending on the VEM approximation order

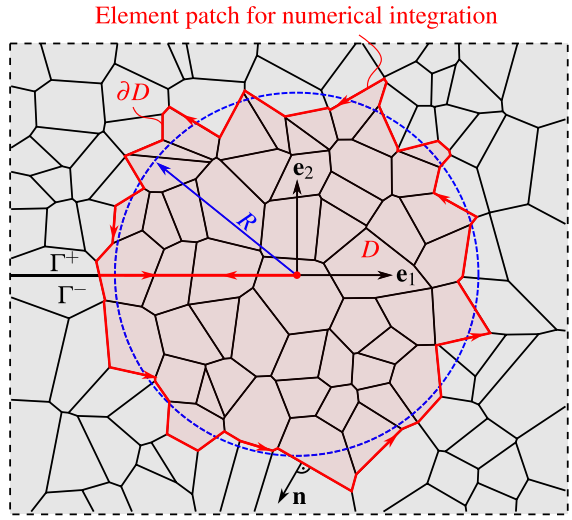

Fig. 5 Arbitrary integration path ∂D for the J-integral around the crack tip

for linear and

$$G_I(t) \approx \frac{1}{2BL}\left[F_2^{\alpha}(t)\Delta u_2^{\alpha-2}(t) + F_2^{\alpha+1}(t)\Delta u_2^{\alpha-1}(t)\right] \tag{64}$$

$$G_{II}(t) \approx \frac{1}{2BL}\left[F_1^{\alpha}(t)\Delta u_1^{\alpha-2}(t) + F_1^{\alpha+1}(t)\Delta u_1^{\alpha-1}(t)\right] \tag{65}$$

for quadratic VEM approximations. Since the difference between the near-field solution for static and dynamic cases is only expressed in the time dependence of the SIFs, Irwin's relationship

$$G(t) = \underbrace{\frac{1+\kappa}{8\mu}K_I^2(t)}_{=\ G_I(t)} + \underbrace{\frac{1+\kappa}{8\mu}K_{II}^2(t)}_{=\ G_{II}(t)} \tag{66}$$

also applies with respect to the crack opening modes for the two-dimensional case.

5.3 J-integral

The path-independent J-integral, which was independently developed by Cherepanov (1967) and Rice (1968) for static problems, is given by

$$\mathbf{J} = B\int_{\partial D} \mathbf{Q}(\mathbf{u})\cdot\mathbf{n}\,\mathrm{d}S + B\int_{D} \rho\,(\nabla\mathbf{u})^{\top}\cdot\ddot{\mathbf{u}}\,\mathrm{d}A \tag{67}$$

in the dynamic case for stationary cracks and without further body forces (Atluri 1982; Nishioka and Atluri 1983b) and dissipation due to material damping. D denotes the integration domain, as illustrated in Fig. 5, and $\mathbf{Q}$ the Eshelby tensor according to Eshelby (1951) or Kuna (2013)

$$\mathbf{Q}(\mathbf{u}) = \psi(\mathbf{u})\mathbf{1} - (\nabla\mathbf{u})^{\top}\cdot\boldsymbol{\sigma}(\mathbf{u})\,. \tag{68}$$

For the elastic energy density the relation

$$\psi(\mathbf{u}) = \frac{1}{2}\,\boldsymbol{\epsilon}(\mathbf{u}):\boldsymbol{\sigma}(\mathbf{u}) = \frac{1}{2}\,\boldsymbol{\epsilon}(\mathbf{u}):\mathbf{C}:\boldsymbol{\epsilon}(\mathbf{u}) \tag{69}$$

holds.

The numerical implementation of integral terms in Eq. (67) requires a VEM-specific modification of the established FEM procedure. To interpolate the displacement field within the element domains, the corresponding projection $\mathbf{u}_{\pi}^{h} = \Pi(\mathbf{u}^{h})$, as introduced in Sec. 3.2, is used instead of element shape functions for both terms.

This results in the approximation

$$\mathbf{J} \approx B \int_{\partial D} \mathbf{Q}(\mathbf{u}_\pi) \cdot \mathbf{n} \, \mathrm{d}S + B \int_{D} \rho \, (\nabla \mathbf{u}_\pi)^\top \cdot \ddot{\mathbf{u}}_\pi \, \mathrm{d}A \tag{70}$$

for the J-integral. This expression can be calculated for any integration path ∂D including the crack tip. In the case of straight crack propagation in the $\mathbf{e}_1$-direction of the crack tip coordinate system, see Fig. 5, the first component of the J-integral is equivalent to the energy release rate. For this work, the numerical integration of Eq. (70) is performed over an element patch, controlled by the radius R around the crack tip. As shown in Fig. 5, all elements whose centroid lies within this radius are assigned to the element patch.

6 Numerical examples

6.1 Griffith crack under impact loading

A Griffith crack under impact loading on the crack surfaces is considered in plane strain conditions. Knowing that the Griffith crack problem covers an infinite domain, the model dimensions in Fig. 6 were chosen for numerical investigations, allowing to neglect size effects. For simplification, the symmetries of the problem were take into account, whereupon a quarter model (blue area in Fig. 6) with corresponding bearings on the symmetry axes was built. The plate has the material properties of steel with a Youngs's modulus of $E = 210$ GPa, a Poisson's ratio of $\nu = 0.3$, and a density of $\rho = 7850$ kg/m^3. The impact load is modelled by a Heaviside step function $H(t)$.

The spatial discretization of the problem employs a mesh of Voronoi cells, shown in Fig. 7. The open-source mesh generator Polymesher, developed by Talischi et al. (2012), was utilized. In order to obtain a high resolution of the stress and displacement field in the vicinity of the crack tip, this region was further refined, exploiting the mesh flexibility of the VEM. The refinement procedure was algorithmically performed by adding vertices to selected element centroids, which were then connected to additional vertices at the element edges. The process was repeated in different stages, allowing a spherical refinement with square elements toward the tip of the crack (see the excerpt in Fig. 7). The properties of the mesh are listed in Table 1. The Hilber-α-Method was utilized for time discretization. For the crack analysis, only the DIM and the MCCI are employed to calculate the relevant Mode I SIF of the underlying problem. The J-integral implementation does not consider crack surface integrals, and is thus not suitable for crack surface loads.

A first numerical investigation of the problem was conducted over the time interval $I_t = [0, t_d]$, for which an analytical solution can be found, e.g., in Freund (1990) or Ravi-Chandar (2004). The characteristic constant t_d represents the time for an elastic dilatation wave to propagate from the crack tip to the left symmetry

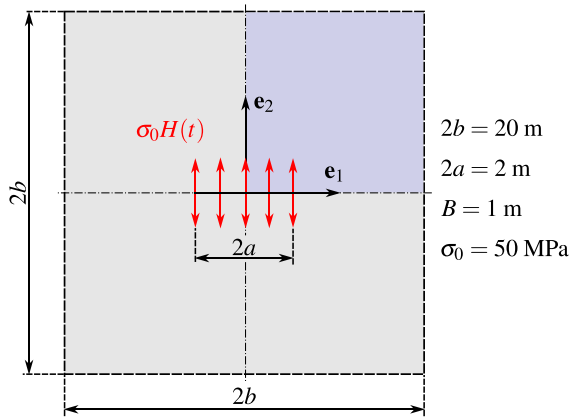

Fig. 6 Griffith crack under impact loading

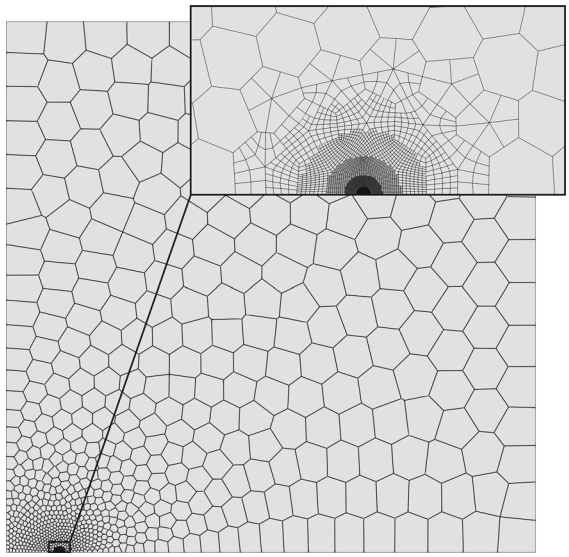

Fig. 7 Voronoi mesh for the quarter model of the plate with square elements at the crack tip

Table 1 Mesh properties for the quarter model of the plate

Property	Value
Number of elements N_E	3944
Max. element diameter $h_{\max}$	≈ 1.074 m
Min. element diameter $h_{\min}$	$\approx 1.2 \cdot 10^{-4}$ m
Ratio L/a	$\approx 1.09 \cdot 10^{-4}$

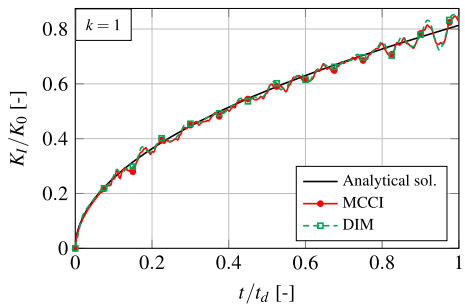

Fig. 8 Normalized SIFs obtained from DIM and MCCI based on a linear VEM approximation, compared to analytical solution

edge and back. With the phase velocity of

$$c_d = \sqrt{\frac{\lambda + 2\mu}{\rho}} \approx 6000 \ \frac{\text{m}}{\text{s}} \tag{71}$$

for a plane strain state, this time is given by

$$t_d = \frac{2a}{c_d} \approx 3.332 \cdot 10^{-4} \ \text{s} \ . \tag{72}$$

The analytical solution then reads

$$K_I(t) = 2\sigma_0 \frac{\sqrt{1-2\nu}}{1-\nu} \sqrt{\frac{c_d t}{\pi}} \ , \tag{73}$$

where the SIF evolves proportional to the square root of time.

Figs. 8 and 9 show the numerical results of the normalized SIF $\left(K_0 = \sigma_0 \sqrt{\pi a}\right)$ for linear ($k = 1$) and quadratic ($k = 2$) approximations, with an increment of $\Delta t = 5 \cdot 10^{-7}$ s and a numerical dissipation rate of $\rho_\infty = 0.98$, compared to the analytical solution. Rayleigh damping was not employed at this point. The SIFs basically follow the analytical solution very well, while the MCCI method is slightly more accurate than the DIM. For both orders of approximation and methods of crack tip loading analyses, the SIFs increase

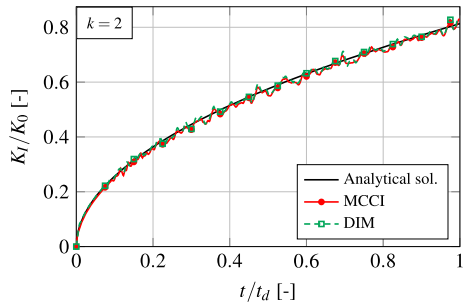

Fig. 9 Normalized SIFs obtained from DIM and MCCI based on a quadratic VEM approximation, compared to analytical solution

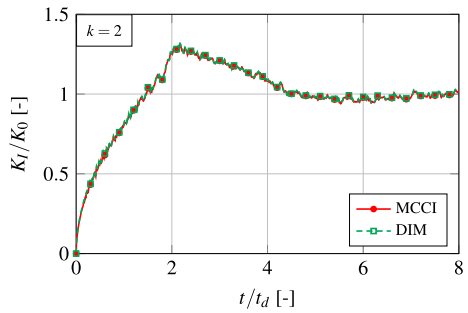

Fig. 10 Normalized SIFs obtained from DIM and MCCI based on a quadratic VEM approximation for a time interval $I_t = [0, 8t_d]$

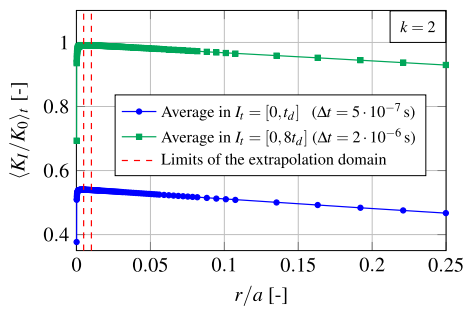

Fig. 11 Normalized SIFs averaged in time intervals I_t, extracted from different DOFs along the crack surface, based on the quadratic VEM

up to approximately $t \approx 0.1t_d$, without noticeable oscillations. The reason for the onset of oscillations at the crack surfaces is that high-frequency components, which are inherent to impact loading, cannot be reproduced by the used time increment. Lower frequency components, however, have a larger effect on crack tip loading, and are thus of major interest from a fracture mechanics point of view. With regard to the order of approximation, quadratic approaches produce less

Table 2 Properties of the additional meshes for the convergence study

Property	Value
Number of elements N_E	806
Max. element diameter $h_{\max}$	≈ 1.074 m
Min. element diameter $h_{\min}$	$\approx 2.68 \cdot 10^{-2}$ m
Ratio L/a	$\approx 2.8 \cdot 10^{-2}$
Number of elements N_E	2549
Max. element diameter $h_{\max}$	≈ 1.074 m
Min. element diameter $h_{\min}$	$\approx 1.51 \cdot 10^{-3}$ m
Ratio L/a	$\approx 1.75 \cdot 10^{-3}$

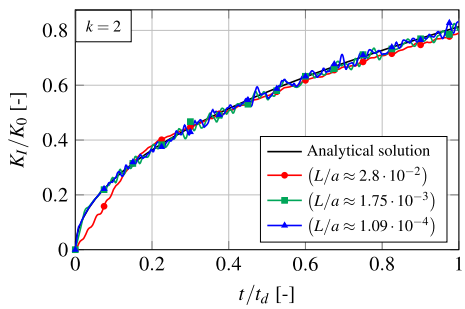

Fig. 12 Convergence study for the normalized SIFs obtained from the DIM based on a quadratic VEM approximation, compared to analytical solution

oscillations compared to a linear interpolation, which becomes particularly evident for $t \to t_d$. It is also observed that the DIM captures fewer frequency components of the solution when using the time averaging approach outlined in Sec. 5.1, resulting in a smoother behavior compared to MCCI.

In Fig. 10, SIFs are depicted in the time interval $I_t = [0, 8t_d]$, maintaining the above damping parameters and adopting a constant time increment of $\Delta t = 2 \cdot 10^{-6}$ s. The crack tip analysis methods reproduce the characteristic behavior of the problem, which is described, e.g., in Freund (1990) or Kuna (2013), very well. After the dilation waves arrive at the crack tip for $t = t_d$, the SIF continues to grow until approximately $K_I(t^\star) \approx 1.3K_0$, which is referred to as dynamic overshoot in the aforementioned literature. The point $t = t^\star$ denotes the time required for Rayleigh waves to reach the crack tip, where they generate compressive stresses, causing the SIF to decrease again. The result is that the SIF finally takes the static value K_0, which is clearly visible in the plot.

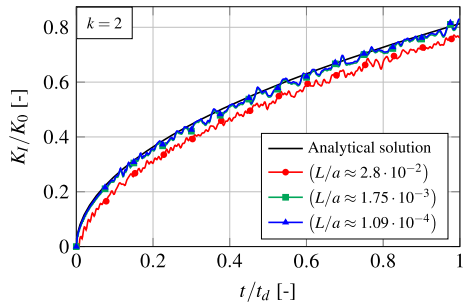

Fig. 13 Convergence study for the normalized SIFs obtained from the MCCI based on a quadratic VEM approximation, compared to analytical solution

For the DIM calculations shown so far, Fig. 11 presents the time averages of the SIFs, extracted from different DOFs along the crack surfaces, for a quadratic VEM approximation. The background is to illustrate the procedure of extrapolation of numerically feasible values to the crack tip at $r/a = 0$, where the SIF is defined from a fracture mechanics point of view. The averaging approach, outlined in Sec. 5.1, obviously provides a similar characteristic curve for the SIF as is known from static calculations. The red dashed lines indicate the limits for the extrapolation domain which was eventually used in each time step.

To complete the example, a convergence study has been conducted for MCCI and DIM, whose accuracy basically depend on the mesh resolution near the crack tip. In addition to the mesh with the properties listed in Table 1, two further discretizations with different refinement stages toward the crack tip are therefore investigated, whereupon two of the coarser concentric meshes of Fig. 7 have been extended to the crack tip. The mesh properties are given in Table 2. Fig. 12 shows the results for the DIM, where the convergence of the method is clearly visible. Even the coarsest mesh, corresponding to the outermost refinement stage of the excerpt in Fig. 7, approximates the analytical solution well, except for $t/t_d < 0.1$. Fig. 13 shows the results for the MCCI, where the SIF is clearly underestimated for the greatest L/a ratio, whereas the $\sqrt{t}$-characteristic is reproduced very well, even for $t/t_d \to 0$. For finer discretizations, the numerical solution converges to the analytical one from below. It is notable here that medium and fine discretizations almost identically reproduce the higher frequency components.

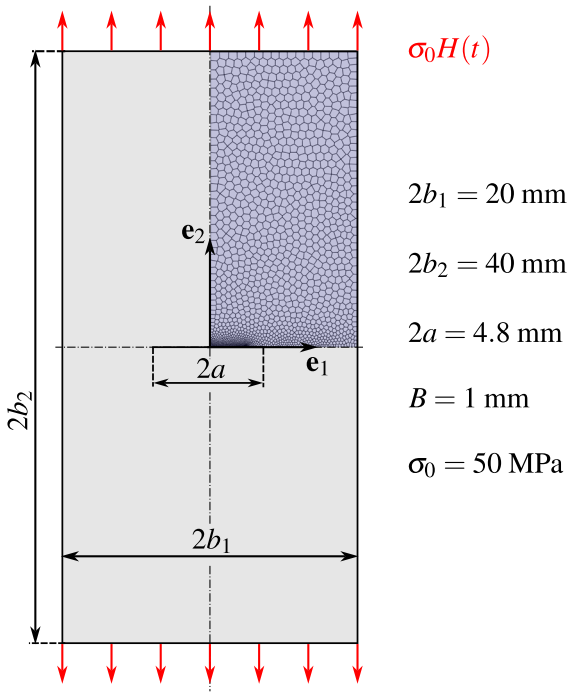

Fig. 14 Central crack in a rectangular plate and Voronoi mesh for the quater model of the plate

Table 3 Properties of the Voronoi mesh of the plate

Property	Value
Number of elements N_E	1750
Max. element diameter $h_{\max}$	≈ 0.820 mm
Min. element diameter $h_{\min}$	$\approx 2.39 \cdot 10^{-3}$ mm
Ratio L/a	$\approx 5.7 \cdot 10^{-4}$

6.2 Central crack in a rectangular plate

For the second numerical example, a central crack in a rectangular plate under plane strain conditions is considered, which is subjected to a remote impact load. Figure 14 illustrates the problem and its geometric characteristics. The material is defined by $\rho = 5000$ kg/m^3, $\mu = 76.92$ GPa and $\nu = 0.3$. Similar to the first example, the impact load on the plate is modelled using a Heaviside step function with $\sigma_0 = 50$ MPa. The time evolution of the relevant SIF K_I has been calculated numerically by Chen (1975), using the Lagrangian FDM and a DIM. This work also presents the velocities of the relevant wave types for the given material parameters, whereby the phase velocity of the dilatation waves is $c_d \approx 7340$ m/s. The problem and the numerical results of Chen (1975) have been addressed in various works on dynamic fracture mechanics, which employed other discretization methods. For instance, Fedelinski et al. (1993) utilized the DBEM to calculate the SIF with the DIM. Enderlein et al. (2003) solved the problem using a linear FEM in connection with the DIM, MCCI, and a J-Integral approach.

To determine the SIF with the VEM, the DIM, the MCCI, and the J-integral are applied. A quarter model (blue area in Fig. 14) was selected based on the symmetry of the problem. For spatial discretization, the Polymesher was used, as in the first example, to generate a mesh with Voronoi cells. The result is also shown in Fig. 14, whereby the relevant mesh properties are provided in Table 3. The Bossak-α-method was employed for time discretization in the range between $t = 0$ s and $T = 15$ μs, with an increment of $\Delta t = 1.5 \cdot 10^{-8}$ s. Damping is not considered in this problem.

Contour diagrams of the calculated displacements u_2 are shown in Fig. 15 for different points in time, whereby the deformation of the plate is magnified by a factor of 250. For $t = 1.875$ μs the crack is still closed, whereas the dilation wave has reached the crack plane in the second sketch. Further crack opening is detected at $t = 7.875$ μs, and the reduction of the local displacement as well as an increasing gradient of u_2 along the horizontal axes indicate that the reflected wave has reached the upper edge. For the last point in time $t = 12.375$ μs, unphysical penetration of the crack faces is observed, since surface contact is not taken into account in the model, allowing for negative values of K_I.

For the application of the J-integral, a study of path independence was first conducted, which is guaranteed by the second integral term in Eq. (70). For this purpose, J-integral curves versus time were generated for different integration radii R using a linear VEM. The results of the study are depicted in Fig. 16 $\left(J_0 = \sigma_0^2 \pi a (1-\nu^2)/E\right)$. It is evident that the J-integral converges for increasing integration radii, confirming path independence and correct implementation. It is remarkable that relatively accurate values are achieved for comparatively small radii with only two or three elements involved in the element patch. In the following calculations, the J-integral values for a radius of $R = 1$ mm were chosen. Fig. 17 shows the element patch near the crack tip for the selected radius.

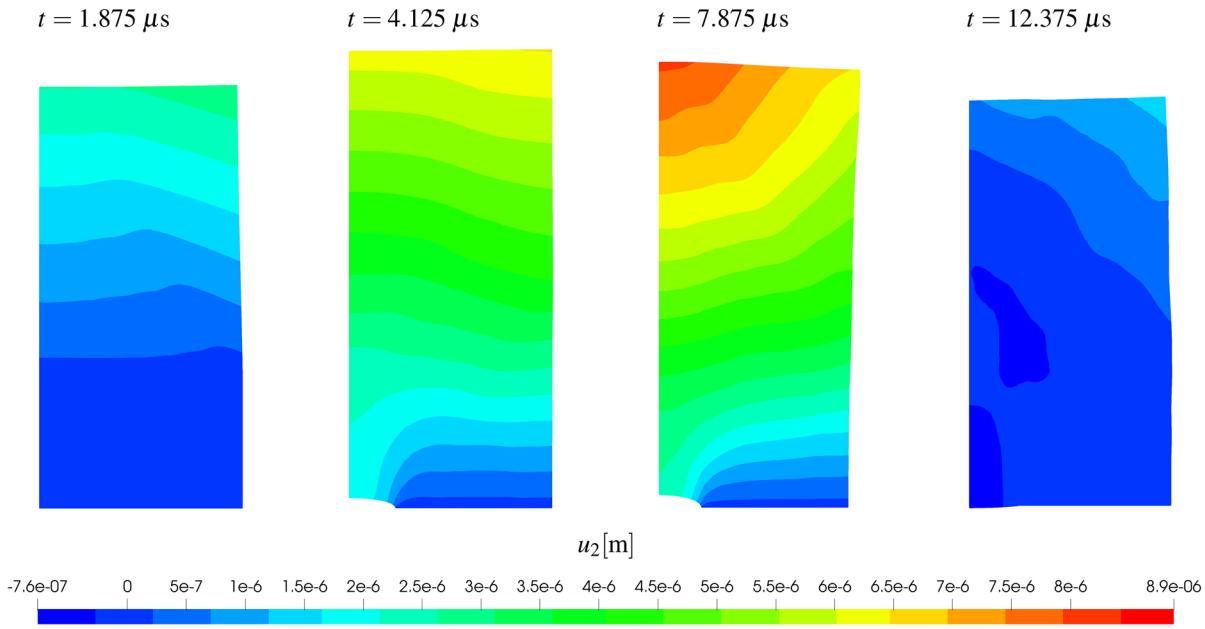

Fig. 15 Displacements u_2 at selected points in time of the problem depicted in Fig. 14 (magnification factor 250)

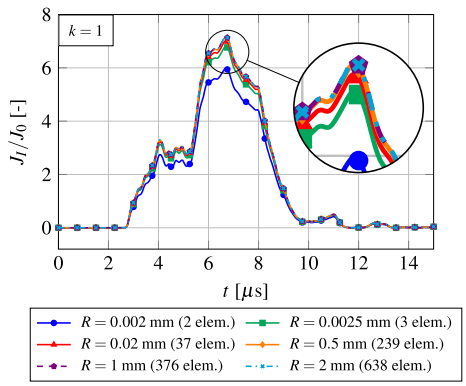

Fig. 16 Normalized J-integral depending on the integration radius R, see Fig. 5, and specification of element numbers per integration patch

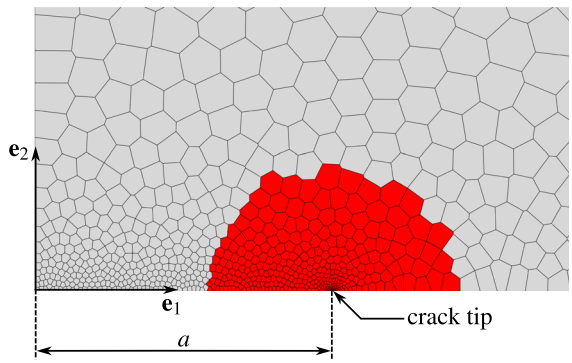

Fig. 17 Element patch (red) near the crack tip for the integration radius $R = 1$ mm of the J-integral

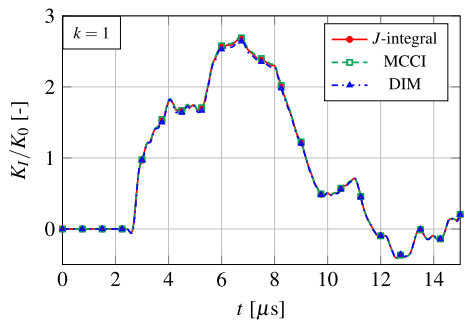

Fig. 18 Comparison of different methods for the calculation of the SIF, based on and a linear VEM approximation

Figs. 18 and 19 illustrate the time evolution of the normalized SIF $\left(K_0 = \sigma_0\sqrt{\pi a}\right)$ for different analysis methods, depending on the VEM approximation in terms of linear ($k = 1$) and quadratic ($k = 2$) elements. The SIF increases significantly up to approximately $t = 7\ \mu s$ as a result of tensile stresses at the crack tip. The magnitude then falls from a maximum of almost three times the approximate static value, and finally becomes negative. For both approximations, the coincidence between the three analysis methods is almost

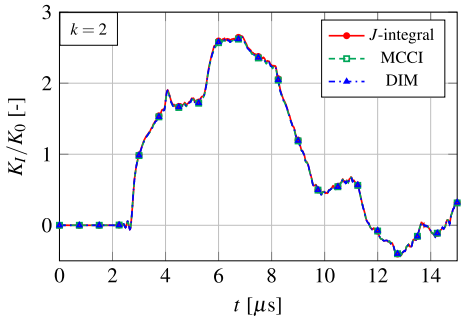

Fig. 19 Comparison of different methods for the calculation of the SIF, based on and a quadratic VEM approximation

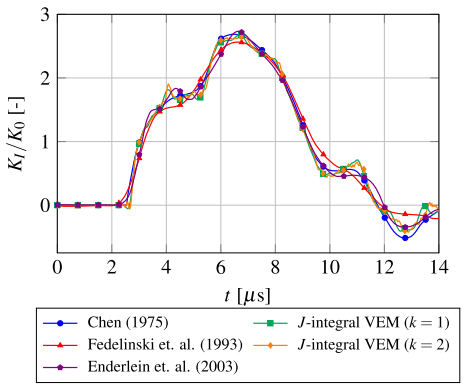

Fig. 20 J-integral based calculation of SIFs for both VEM approximation orders and reference solutions

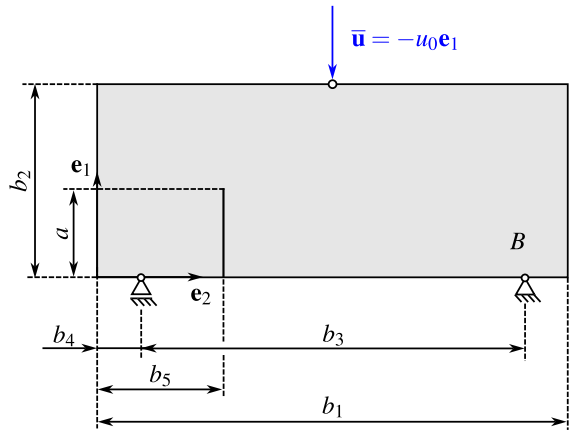

Fig. 21 Model of the SENB specimen with boundary conditions

Table 4 Geometric characteristics of the SENB specimen

Quantity	Value
Length b_1	20 mm
Length b_2	9 mm
Length b_3	16 mm
Length b_4	2 mm
Length b_5	6 mm
Crack length a	4 mm
Thickness B	1 mm

perfect, in particular for $k = 2$. Quadratic approximations, however, produce more oscillation by capturing higher frequencies in the system.

The J-integrals for both approximation orders exhibit comparable characteristics when benchmarked against the numerical results of Chen (1975); Fedelinski et al. (1993), and the J-integral from Enderlein et al. (2003), which are all together depicted in Fig. 20. However, the VEM solutions exhibit some more oscillations compared to the other calculations.

6.3 Three-point bending test with single-edge notched bend (SENB) specimen

For the last numerical example, the mixed mode problem of a three-point bending test with a non-central notch in an SENB specimen is studied. Fig. 21 shows the underlying problem, while the geometric model characteristics are depicted in Table 4. The specimen is made of aluminium with Youngs's modulus $E = 70$ GPa, Poisson's ratio $\nu = 0.35$ and density of $\rho = 2700$ kg/m^3. Furthermore, the specimen is assumed in a plane stress state.

The boundary condition acting on top of the specimen follows a ramp function with

$$u_0 = \begin{cases} \dfrac{\hat{u}}{\tau} t & , \quad 0 \leq t < \tau \\ \hat{u} & , \quad \tau \leq t \leq T \end{cases} , \tag{74}$$

which is defined by the maximum value $\hat{u}$ and the time constant τ. A mesh of quadrilateral elements was chosen for spatial discretization, with refinement near the crack tip using a Quadtree structure with hanging nodes, see Fig. 22 and Table 5. The SIFs are calculated with the DIM and the Bossak-α-Method in the range between $t = 0$ s and $T = 15$ μs. The MCCI and the J-integral according to Eq. (67) are not applicable, due to damping, investigated in the following.

A first numerical investigation of the problem is conducted with load parameters $\hat{u} = 1 \cdot 10^{-6}$ m and

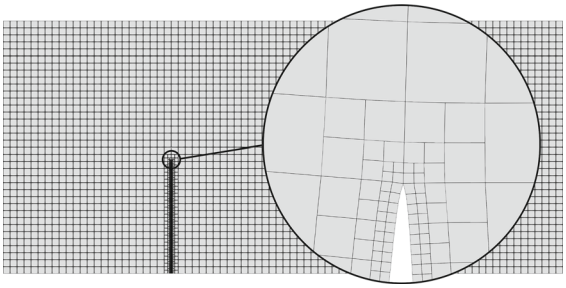

Fig. 22 Mesh of the SENB specimen model with an excerpt of the opened crack faces

Table 5 Mesh properties for the SENB specimen

Property	Value
Number of elements N_E	4344
Max. element diameter $h_{\max}$	≈ 0.354 mm
Min. element diameter $h_{\min}$	≈ 0.0221 mm

$\tau = 0.1$ μs, and different values of the Rayleigh damping parameter α. An adaptive increment is chosen for time discretization, whereby the time interval, in which the displacement load increases, is discretized with $\Delta t = \tau/10$. The remaining time interval is calculated with an increment of $\Delta t = 2.5 \cdot 10^{-8}$ s. Fig. 23 shows the evolutions over time of the SIFs for a VEM approximation order of $k = 2$. Negative values for K_I are associated with unphysical penetration of the crack faces. While the SIFs exhibit strong oscillations in the absence of damping, the oscillations reduce for increasing values of the damping parameter. For strong stiffness-proportional damping with $\alpha = 1 \cdot 10^{-3}$ s, the SIFs rapidly take on their static value.

With identical spatial and time discretization parameters, as well as the same settings for the boundary conditions, a study on numerical damping is also conducted. The SIFs, depending on the algorithmic damping parameter ρ_∞, are depicted in Fig. 24. Comparing both types of damping, it is evident that numerical damping only affects higher, non-energy-rich modes of the system, while Rayleigh damping changes the system behavior by dissipating the energy $E_{\text{diss}}(t_n) = \mathbf{u}^\top(t_n)\mathbf{D}\dot{\mathbf{u}}(t_n)$ per time step.

Without any influence of damping, the SIFs are investigated for various ramp parameters τ and with linear VEM approximation order ($k = 1$). A constant time increment $\Delta t = 1 \cdot 10^{-8}$ s has been chosen for this study. The results are illustrated in Fig. 25. As expected, the maxima are reduced with increasing τ, coming along with a smoother and less energetic impact. Furthermore, oscillations are significantly more pronounced for smaller values of τ. Apparently, a strong impact stimulates higher order modes, which is well-known from signal theory.

To verify the mixed-mode problem, a FEM calculation was finally performed for two selected ramp parameters using the commercial solver Abaqus/Standard on a typical FE mesh with $N_E = 15012$ quadrilateral elements. Fig. 26 shows the results. The dashed curves represent the FEM and correspond very well with the VEM results. The comparison demon-

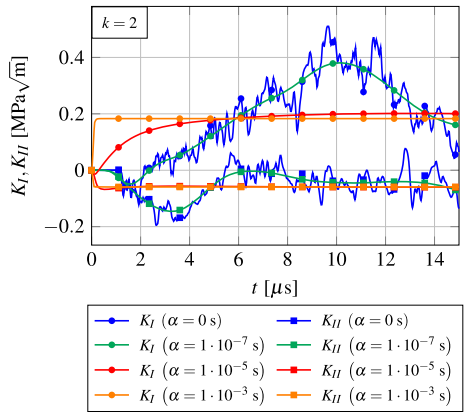

Fig. 23 SIFs obtained from the DIM, based on quadratic VEM approximation, depending on the Rayleigh damping parameter α

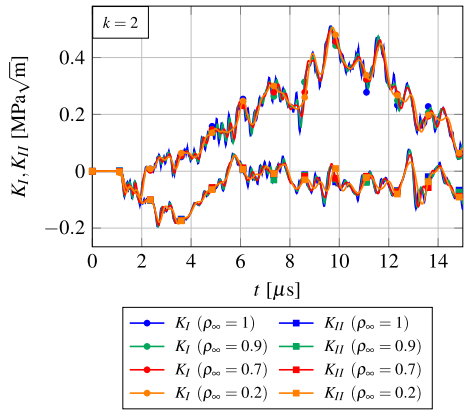

Fig. 24 SIFs obtained from the DIM, based on quadratic VEM approximation, depending on the algorithmic dissipation parameter ρ_∞

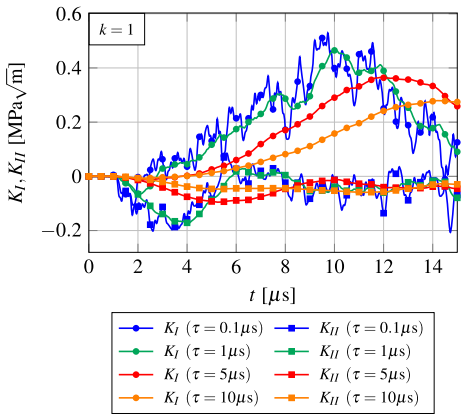

Fig. 25 SIFs obtained from the DIM, based on linear VEM approximation, depending on the ramp parameter τ

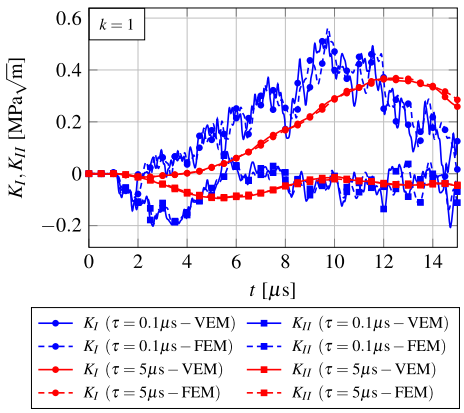

Fig. 26 SIFs obtained from the DIM, based on linear VEM and FEM approximations, depending on the ramp parameter τ

strates the accuracy of the VEM calculations also in the mixed-mode case.

7 Summary and conclusion

The application of the VEM for dynamic crack loading analysis is presented. For this purpose, the spatial and time discretization of the underlying IBVP is discussed. After that, the fracture mechanics quantities and their numerical implementation used for the analysis are outlined. In particular, the realization of a dynamic J-integral is illustrated in the context of the VEM. Three numerical examples are presented. For the first and second example, which are mode-I loaded, the results are compared with findings from the literature, applying the DIM and the MCCI, and for the second example also the J-integral. For the last mixed mode problem, which is verified by means of in-house FEM calculations, a study of the damping parameters and the time constant of the displacement-controlled impact are conducted with the DIM.

The results demonstrate that the VEM is suitable for applications in the field of dynamic fracture mechanics, and constitutes an alternative compared to established approaches, such as the FEM or the BEM. Even though the mathematical description is extensive, the method straightforwardly allows for a high mesh flexibility in terms of, inter alia, arbitrarily shaped convex and non-convex elements, hanging nodes and strong mesh gradations. Established methods of dynamic crack tip loading analysis are applicable within the VEM context, although special treatment is required, e.g., due to the lack of element shape functions. All these properties are supposed to make the VEM highly suitable in the context of simulating dynamic crack propagation for future works.

These should take into account crack face contact, including friction, to avoid negative K_I, and to appropriately calculate K_{II} and thus crack deflection angles. Virtual elements with high order approximations, i.e. $k > 2$, are pursued in future work, which are particularly suitable to resolve wave fronts initiated by impact loading. Finally, a 3D implementation of the dynamic VEM is desirable to enable crack growth simulations with mode-III contribution.

Author contributions Philipp Wappler: Implementation, Validation, Preparation of results, Visualization, Writing – Kevin Schmitz: Implementation, Validation, Supervision – Andreas Ricoeur: Writing, Supervision, Review.

Funding Open Access funding enabled and organized by Projekt DEAL.

Data Availability Statement No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's

Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Ahmad B, Alsaedi A, Brezzi F, Marini L, Russo A (2013) Equivalent projectors for virtual element methods. Comput Math Appl 66(3):376–391. https://doi.org/10.1016/j.camwa.2013.05.015

Aldakheel F, Hudobivnik B, Wriggers P (2019) Virtual element formulation for phase-field modeling of ductile fracture. Int J Mult Comp Eng 17(2):181–200. https://doi.org/10.1615/IntJMultCompEng.2018026804

Aliabadi MH (1997) Boundary Element Formulations in Fracture Mechanics. Appl Mech Rev 50(2):83–96. https://doi.org/10.1115/1.3101690

Antonietti PF (2022) The virtual element method and its applications, vol 31 in SEMA SIMAI Springer Series, 1st ed. Springer International Publishing AG, Cham

Antonietti PF, Manzini G, Mazzieri I, Mourad HM, Verani M (2021) The arbitrary-order virtual element method for linear elastodynamics models: convergence, stability and dispersion-dissipation analysis. Numer Meth Eng 122(4):934–971. https://doi.org/10.1002/nme.6569

Artioli E, Beirão Da Veiga L, Lovadina C, Sacco E (2017) Arbitrary order 2D virtual elements for polygonal meshes: part I, elastic problem. Comput Mech 60(3):355–377 https://doi.org/10.1007/s00466-017-1404-5

Atluri SN (1982) Path-independent integrals in finite elasticity and inelasticity, with body forces, inertia, and arbitrary crack-face conditions. Eng Fract Mech 16(3):341–364. https://doi.org/10.1016/0013-7944(82)90113-8

Bathe KJ (1996) Finite element procedures. Prentice Hall, Englewood Cliffs

Beirão da Veiga L, Brezzi F, Cangiani A, Manzini G, Marini LD, Russo A (2013) Basic principles of virtual element methods. Math Models Methods Appl Sci 23(01):199–214. https://doi.org/10.1142/S0218202512500492

Beirão da Veiga L, Brezzi F, Marini LD (2013) Virtual elements for linear elasticity problems. SIAM J Numer Anal 51(2):794–812. https://doi.org/10.1137/120874746

Beirão da Veiga L, Brezzi F, Marini LD, Russo A (2014) The Hitchhiker's guide to the virtual element method. Math Models Methods Appl Sci 24(08):1541–1573. https://doi.org/10.1142/S021820251440003X

Beirão da Veiga L, Brezzi F, Marini LD, Russo A (2023) The virtual element method. Acta Numer 32:123–202. https://doi.org/10.1017/S0962492922000095

Belytschko T, Black T (1999) Elastic crack growth in finite elements with minimal remeshing. Int J Numer Meth Eng 45(5):601–620

Buchholz FG (1984) Improved formulae for the finite element calculation of the strain energy release rate by modified crack closure integral method. Accuracy, reliability and training in fem technology. Robinson and Associates, Dorset, pp 650–659

Chen YM (1975) Numerical computation of dynamic stress intensity factors by a Lagrangian finite-difference method (the HEMP code). Eng Fract Mech 7:653–660

Chen Y, Sun D, Perego U, Li Q (2025) Brittle crack propagation simulation based on the Virtual Element Method and J_k -integral fracture criterion. Eng Fract Mech 314:110684. https://doi.org/10.1016/j.engfracmech.2024.110684

Cherepanov G (1967) Rasprostranenie trechin v sploshnoi srede (About crack advance in the continuum). Prikladnaja Matematika i Mekhanica 31:478–488

Chin EB, Sukumar N (2021) Scaled boundary cubature scheme for numerical integration over planar regions with affine and curved boundaries. Comput Methods Appl Mech Eng 380:113796. https://doi.org/10.1016/j.cma.2021.113796

Chirino F, Dominguez J (1989) Dynamic analysis of cracks using boundary element method. Eng Fract Mech 34(5–6):1051–1061. https://doi.org/10.1016/0013-7944(89)90266-X

Choi H, Chi H, Park K (2023) Virtual element method for mixed-mode cohesive fracture simulation with element split and domain integral. Int J Fract 240(1):51–70. https://doi.org/10.1007/s10704-022-00675-7

Chung J, Hulbert GM (1993) A time integration algorithm for structural dynamics with improved numerical dissipation: the generalized-alpha-method. J Appl Mech 60(2):371–375. https://doi.org/10.1115/1.2900803

Cihan M, Hudobivnik B, Aldakheel F, Wriggers P (2021) Virtual element formulation for finite strain elastodynamics. Comput Model Eng Sci 129(3):1151–1180

Dassi F (2025) Vem++, a C++ library to handle and play with the virtual element method. Numer Algor. https://doi.org/10.1007/s11075-025-02059-z

Enderlein M, Ricoeur A, Kuna M (2003) Comparison of finite element techniques for 2D and 3D crack analysis under impact loading. Int J Solids Struct 40(13–14):3425–3437. https://doi.org/10.1016/S0020-7683(03)00117-3

Eshelby J (1951) The force on an elastic singularity. Phil Trans R Soc Lond A 244(877):87–112. https://doi.org/10.1098/rsta.1951.0016

Fedelinski P, Aliabadi M, Rooke D (1993) The dual boundary element method in dynamic fracture mechanics. Eng Anal Boundary Elem 12(3):203–210. https://doi.org/10.1016/0955-7997(93)90016-E

Fedelinski P, Aliabadi MH, Rooke DP (1997) The time-domain DBEM for rapidly growing cracks. Int J Numer Meth Eng 40(9):1555–1572

Freund LB (1990) Dynamic fracture mechanics, 1st ed. Cambridge University Press. https://doi.org/10.1017/CBO9780511546761. https://www.cambridge.org/core/product/identifier/9780511546761/type/book

Gross D, Seelig T (2018) Fracture mechanics: with an introduction to micromechanics, 3rd ed. Mechanical Engineering Series, Springer International Publishing : Imprint: Springer, Cham. https://doi.org/10.1007/978-3-319-71090-7

Gürses E, Miehe C (2009) A computational framework of three-dimensional configurational-force-driven brittle crack propagation. Comput Methods Appl Mech Eng 198(15–16):1413–1428. https://doi.org/10.1016/j.cma.2008.12.028

Hilber HM, Hughes TJR, Taylor RL (1977) Improved numerical dissipation for time integration algorithms in structural dynamics. Earthq Eng Struct Dyn 5(3):283–292. https://doi.org/10.1002/eqe.4290050306

Hughes TJR (ed) (2000) The finite element method: linear static and dynamic finite element analysis. Dover Publications, Mineola

Huo S, Liu G, Zhang J, Song C (2020) A smoothed finite element method for octree-based polyhedral meshes with large number of hanging nodes and irregular elements. Comput Methods Appl Mech Eng 359:112646. https://doi.org/10.1016/j.cma.2019.112646

Hussein A, Aldakheel F, Hudobivnik B, Wriggers P, Guidault PA, Allix O (2019) A computational framework for brittle crack-propagation based on efficient virtual element method. Finite Elem Anal Des 159:15–32. https://doi.org/10.1016/j.finel.2019.03.001

Jih C, Sun C (1990) Evaluation of a finite element based crack-closure method for calculating static and dynamic strain energy release rates. Eng Fract Mech 37(2):313–322. https://doi.org/10.1016/0013-7944(90)90043-G

Kuna M (2013) Finite elements in fracture mechanics: theory—numerics—applications. Solid mechanics and its applications. Springer, Netherlands, Dordrecht. https://doi.org/10.1007/978-94-007-6680-8

Kus P, Solin P, Andrs D (2014) Arbitrary-level hanging nodes for adaptive h p -FEM approximations in 3D. J Comput Appl Math 270:121–133. https://doi.org/10.1016/j.cam.2014.02.010

Leng Y, Svolos L, Boureima I, Manzini G, Plohr JN, Mourad HM (2025) Arbitrary order virtual element methods for high-order phase-field modeling of dynamic fracture. Numer Meth Eng 126(1):e7605. https://doi.org/10.1002/nme.7605

Liguori F, Madeo A, Marfia S, Garcea G, Sacco E (2024) A stabilization-free hybrid virtual element formulation for the accurate analysis of 2D elasto-plastic problems. Comput Methods Appl Mech Eng 431:117281. https://doi.org/10.1016/j.cma.2024.117281

Marfia S, Monaldo E, Sacco E (2022) Cohesive fracture evolution within virtual element method. Eng Fract Mech 269:108464. https://doi.org/10.1016/j.engfracmech.2022.108464

Mengolini M, Benedetto MF, Aragón AM (2019) An engineering perspective to the virtual element method and its interplay with the standard finite element method. Comput Methods Appl Mech Eng 350:995–1023. https://doi.org/10.1016/j.cma.2019.02.043

Moës N, Dolbow J, Belytschko T (1999) A finite element method for crack growth without remeshing. Int J Numer Meth Engng 46(1):131–150

Nakamura T, Shih CF, Freund LB (1985) Computational methods based on an energy integral in dynamic fracture. Int J Fract 27(3–4):229–243. https://doi.org/10.1007/BF00017970

Newmark NM (1959) A method of computation for structural dynamics. J Engrg Mech Div 85(3):67–94. https://doi.org/10.1061/JMCEA3.0000098

Nguyen-Thanh VM, Zhuang X, Nguyen-Xuan H, Rabczuk T, Wriggers P (2018) A virtual element method for 2D linear elastic fracture analysis. Comput Methods Appl Mech Eng 340:366–395. https://doi.org/10.1016/j.cma.2018.05.021

Nishioka T, Atluri S (1983) A numerical study of the use of path independent integrals in elasto-dynamic crack propagation. Eng Fract Mech 18(1):23–33. https://doi.org/10.1016/0013-7944(83)90092-9

Nishioka T, Atluri S (1983) Path-independent integrals, energy release rates, and general solutions of near-tip fields in mixed-mode dynamic fracture mechanics. Eng Fract Mech 18(1):1–22. https://doi.org/10.1016/0013-7944(83)90091-7

Park K, Chi H, Paulino GH (2019) On nonconvex meshes for elastodynamics using virtual element methods with explicit time integration. Comput Methods Appl Mech Eng 356:669–684

Perl M, Shmuely M, Peretz D (1982) An improved dynamic crack propagation simulation in the SEN specimen by the SMF2D code. Int J Fract 19(1):17–27. https://doi.org/10.1007/BF00012489

Rashid M (1998) The arbitrary local mesh replacement method: an alternative to remeshing for crack propagation analysis. Comput Methods Appl Mech Eng 154(1–2):133–150. https://doi.org/10.1016/S0045-7825(97)00068-6

Ravi-Chandar K (ed) (2004) Dynamic fracture. Elsevier, Amsterdam

Rice JR (1968) A path independent integral and the approximate analysis of strain concentration by notches and cracks. J Appl Mech 35(2):379–386. https://doi.org/10.1115/1.3601206

Shmuely M, Alterman ZS (1973) Crack propagation analysis by finite differences. J Appl Mech 40(4):902–908. https://doi.org/10.1115/1.3423185

Sinha AK, Pamnani G (2025) Applications of XFEM in fracture analysis: a review. Iran J Sci Technol Trans Civ Eng. https://doi.org/10.1007/s40996-025-01955-7

Slaughter WS (2002) The linearized theory of elasticity. Birkhäuser Boston, Boston. https://doi.org/10.1007/978-1-4612-0093-2

Song C, Wolf JP (1996) Consistent infinitesimal finite-element cell method: three-dimensional vector wave equation. Int J Numer Meth Engng 39(13):2189–2208

Song C, Wolf JP (1997) The scaled boundary finite-element method-alias consistent infinitesimal finite-element cell method-for elastodynamics. Comput Methods Appl Mech Eng 147(3–4):329–355. https://doi.org/10.1016/S0045-7825(97)00021-2

Sukumar N, Tupek MR (2022) Virtual elements on agglomerated finite elements to increase the critical time step in elastodynamic simulations. Numer Meth Eng 123(19):4702–4725. https://doi.org/10.1002/nme.7052

Talischi C, Paulino GH, Pereira A, Menezes IFM (2012) PolyMesher: a general-purpose mesh generator for polygonal elements written in Matlab. Struct Multidisc Optim 45(3):309–328. https://doi.org/10.1007/s00158-011-0706-z

Wen PH, Aliabadi MH, Rooke DP (1996) Application of the weight function method to two-dimensional elastodynamic fracture mechanics. Int J Fract 76(3):193–206. https://doi.org/10.1007/BF00048286

Wood WL, Bossak M, Zienkiewicz OC (1980) An alpha modification of Newmark's method. Numer Meth

Eng 15(10):1562–1566. https://doi.org/10.1002/nme.1620151011

Wriggers P, Hudobivnik B (2017) A low order virtual element formulation for finite elasto-plastic deformations. Comput Methods Appl Mech Eng 327:459–477. https://doi.org/10.1016/j.cma.2017.08.053

Wriggers P, Aldakheel F, Hudobivnik B (2024) Virtual element methods in engineering sciences. Springer International Publishing, Cham. https://doi.org/10.1007/978-3-031-39255-9

Xu BB, Peng F, Junker P, Wriggers P (2025) Virtual element method with non-matching and adaptive meshes for phase field fracture. Comput Mech. https://doi.org/10.1007/s00466-025-02665-7

Yang Z (2006) Application of scaled boundary finite element method in static and dynamic fracture problems. Acta Mech Mech Sinica 22(3):243–256. https://doi.org/10.1007/s10409-006-0110-x

Zhang P, Du C, Zhao W, Sun L (2021) Dynamic crack face contact and propagation simulation based on the scaled boundary finite element method. Comput Methods Appl Mech Eng 385:114044. https://doi.org/10.1016/j.cma.2021.114044

Zienkiewicz OC, Taylor RL (2000) The finite element method, 5th edn. Butterworth-Heinemann, Oxford, Boston

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.